Revision 2021 • Diploma in Mechanical Engineering • Semester 4

Thermal Engineering

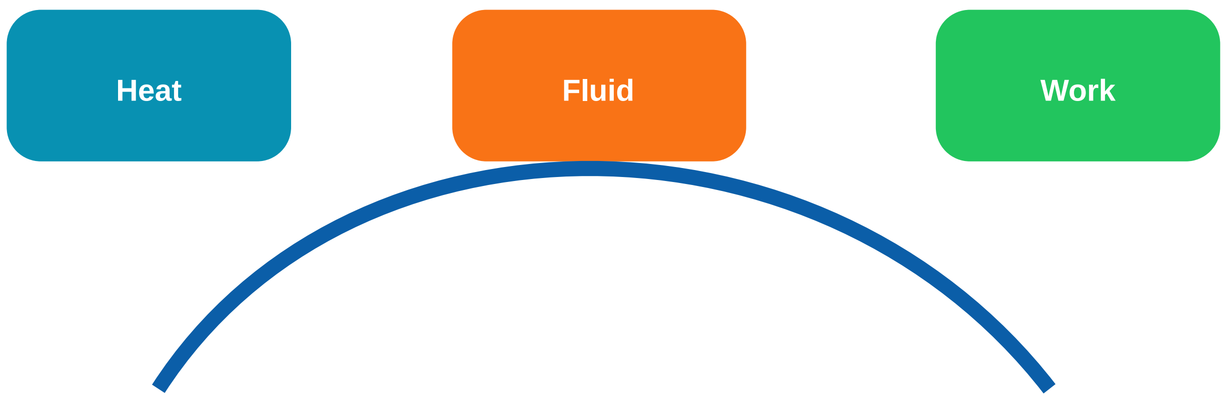
A complete professional handbook for Course 4021. It covers thermodynamics, gas laws, air-standard cycles, IC engine performance, steam, boilers, turbines, heat transfer, heat exchangers and compressors with diagrams, formulas, solved examples, practice questions, model paper and answer guide.

Code
4021

Credits
4

Periods
60

Category
Program Core



Energy conversion and heat-flow thinking

Course Overview

Item	Value

> Diagram + formula + unit + explanation = scoring method.

Learning Roadmap

CO1 • 16 hrs

Thermodynamics, laws and processes

scope and application, system, boundary, surrounding, pressure, temperature, enthalpy, Boyle, Charles, Joule and Avogadro laws, gas equation and Mayer relation

Draw: P-V/T-S diagrams, machine line diagrams and flow charts.
Calculate: formula-based simple problems.

CO2 • 15 hrs

Air-standard cycles and IC engine testing

air-standard assumptions, Carnot, Otto and Diesel cycles, P-V and T-S diagrams, two-stroke and four-stroke engines, SI and CI engines

Draw: P-V/T-S diagrams, machine line diagrams and flow charts.
Calculate: formula-based simple problems.

CO3 • 15 hrs

Steam, boilers and turbines

steam formation, wet, dry and superheated steam, steam tables, Mollier chart, throttling process

Draw: P-V/T-S diagrams, machine line diagrams and flow charts.
Calculate: formula-based simple problems.

CO4 • 11 hrs

Heat transfer, heat exchangers and compressors

conduction, convection and radiation, Fourier law, plane wall conduction, radiation terms, black and grey body

Draw: P-V/T-S diagrams, machine line diagrams and flow charts.
Calculate: formula-based simple problems.

CO1 • 16 hours

1. Thermodynamics, laws and processes

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

Chapter explanation

Study this module as an engineering system: identify energy input, output, working substance, losses and practical limitation. Write answers using definition, labelled diagram, working, formula, application and common mistake.

Thermodynamics-ൽ system boundary, property, heat, work എന്നിവ വേർതിരിച്ച് പഠിക്കണം. Temperature kelvin ആക്കാതെ gas equation ഉപയോഗിക്കരുത്.

Core topics

scope and application · **system, boundary, surrounding** · **pressure, temperature, enthalpy** · **Boyle, Charles, Joule and Avogadro laws** · **gas equation and Mayer relation** · **heat, work and equilibrium** · **thermodynamic processes** · **laws of thermodynamics** · **entropy and reversibility** · **P-V and T-S diagrams**

Diagram lab

Thermodynamics, laws and processes

Common mistakes

- Formula without units.
- Unlabelled diagrams.
- Wrong state or process assumption.
- Skipping application and limitation.

Expected questions

1. Explain scope and application with diagram / formula / example.
2. Explain system, boundary, surrounding with diagram / formula / example.

3. Explain pressure, temperature, enthalpy with diagram / formula / example.
4. Explain Boyle, Charles, Joule and Avogadro laws with diagram / formula / example.
5. Explain gas equation and Mayer relation with diagram / formula / example.
6. Explain heat, work and equilibrium with diagram / formula / example.
7. Explain thermodynamic processes with diagram / formula / example.

CO2 • 15 hours

2. Air-standard cycles and IC engine testing

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

Chapter explanation

Study this module as an engineering system: identify energy input, output, working substance, losses and practical limitation. Write answers using definition, labelled diagram, working, formula, application and common mistake.

> IC engine testing-ൽ BP shaft output ആണ്, IP cylinder power ആണ്, difference friction power ആണ്.

Core topics

air-standard assumptions | Carnot, Otto and Diesel cycles | P-V and T-S diagrams | two-stroke and four-stroke engines | SI and CI engines | valve timing and port timing | IP, BP and FP | thermal efficiencies | SFC, Morse test and heat balance

Diagram lab

Air-standard cycles and IC engine testing

Common mistakes

- Formula without units.
- Unlabelled diagrams.
- Wrong state or process assumption.
- Skipping application and limitation.

Expected questions

1. Explain air-standard assumptions with diagram / formula / example.
2. Explain Carnot, Otto and Diesel cycles with diagram / formula / example.
3. Explain P-V and T-S diagrams with diagram / formula / example.
4. Explain two-stroke and four-stroke engines with diagram / formula / example.
5. Explain SI and CI engines with diagram / formula / example.
6. Explain valve timing and port timing with diagram / formula / example.
7. Explain IP, BP and FP with diagram / formula / example.

CO3 • 15 hours

3. Steam, boilers and turbines

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

Chapter explanation

Study this module as an engineering system: identify energy input, output, working substance, losses and practical limitation. Write answers using definition, labelled diagram, working, formula, application and common mistake.

> Steam table question-ൽ state first identify ചെയ്യുക. Wet steam ആണെങ്കിൽ h=hf+xhfg ഉപയോഗിക്കുക.

Core topics

steam formation · wet, dry and superheated steam · steam tables · Mollier chart · throttling process · boiler classification · Cochran boiler · Babcock and Wilcox boiler · mountings and accessories · impulse and reaction turbines

Diagram lab

Steam, boilers and turbines

Common mistakes

- Formula without units.
- Unlabelled diagrams.
- Wrong state or process assumption.
- Skipping application and limitation.

Expected questions

1. Explain steam formation with diagram / formula / example.
2. Explain wet, dry and superheated steam with diagram / formula / example.
3. Explain steam tables with diagram / formula / example.
4. Explain Mollier chart with diagram / formula / example.
5. Explain throttling process with diagram / formula / example.
6. Explain boiler classification with diagram / formula / example.
7. Explain Cochran boiler with diagram / formula / example.

CO4 • 11 hours

4. Heat transfer, heat exchangers and compressors

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

Chapter explanation

Study this module as an engineering system: identify energy input, output, working substance, losses and practical limitation. Write answers using definition, labelled diagram, working, formula, application and common mistake.

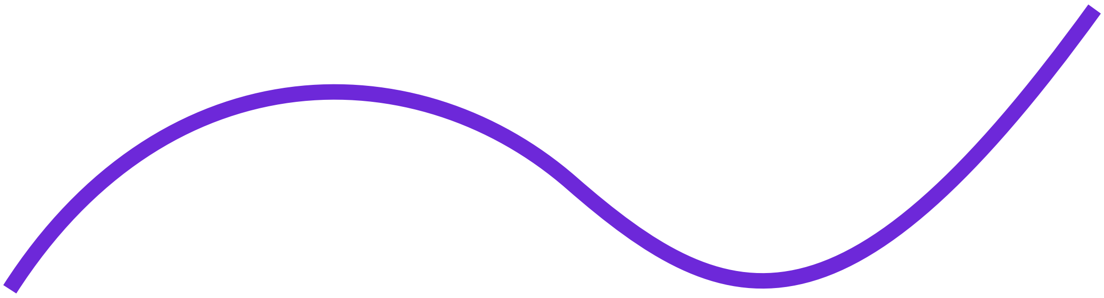
> Heat transfer-ൽ mode identify ചെയ്യണം. Conduction solid wall, convection fluid motion, radiation no medium.

Core topics

- conduction, convection and radiation
- Fourier law
- plane wall conduction
- radiation terms
- black and grey body
- Newton-Richman law
- heat exchanger classification
- effectiveness
- reciprocating compressor
- rotary compressor types

Diagram lab

Course 4021



Heat transfer, heat exchangers and compressors

Common mistakes

- Formula without units.
- Unlabelled diagrams.
- Wrong state or process assumption.
- Skipping application and limitation.

Expected questions

1. Explain conduction, convection and radiation with diagram / formula / example.
2. Explain Fourier law with diagram / formula / example.
3. Explain plane wall conduction with diagram / formula / example.
4. Explain radiation terms with diagram / formula / example.
5. Explain black and grey body with diagram / formula / example.
6. Explain Newton-Richman law with diagram / formula / example.
7. Explain heat exchanger classification with diagram / formula / example.

Formula Bank

pV=mRT

> $pV=mRT$ - ideal gas equation

Write variables, units, substitution and final unit in exam.

$C_p - C_v = R$

$C_p - C_v = R$ - Mayer relation

Write variables, units, substitution and final unit in exam.

$W = p(V_2 - V_1)$

$W = p(V_2 - V_1)$ - constant pressure work

Write variables, units, substitution and final unit in exam.

$pV^\gamma = \text{constant}$

$pV^\gamma = \text{constant}$ - adiabatic process

Write variables, units, substitution and final unit in exam.

$\eta_{Otto} = 1 - 1/r^{(\gamma-1)}$

$\eta_{Otto} = 1 - 1/r^{(\gamma-1)}$ - Otto cycle efficiency

Write variables, units, substitution and final unit in exam.

$\eta_m = BP/IP$

$\eta_m = BP/IP$ - mechanical efficiency

Write variables, units, substitution and final unit in exam.

SFC=fuel used per BP hour

> SFC=fuel used per BP hour

Write variables, units, substitution and final unit in exam.

h=hf+xhfg

> h=hf+xhfg - wet steam enthalpy

Write variables, units, substitution and final unit in exam.

Q=kA(T1-T2)/L

> Q=kA(T1-T2)/L - Fourier law

Write variables, units, substitution and final unit in exam.

effectiveness=actual heat transfer/maximum possible

> effectiveness=actual heat transfer/maximum possible

Write variables, units, substitution and final unit in exam.

Solved Examples

Gas equation

Question: $p=200$ kPa, $V=0.5\ m^3$, $T=300$ K, $R=0.287$. Find mass.

$m=pV/RT=200\times0.5/(0.287\times300)=1.16$ kg.

Exam tip: Given data → formula → substitution → answer with unit.

Otto efficiency

Question: $r=8$ and $\gamma=1.4$.

$\eta=1-1/8^{0.4}=56.5\%$.

Exam tip: Given data → formula → substitution → answer with unit.

Engine powers

Question: $IP=25$ kW, $BP=20$ kW.

$FP=5$ kW and $\eta m=80\%$.

Exam tip: Given data → formula → substitution → answer with unit.

Wet steam

Question: $hf=640$, $hfg=2100$, $x=0.9$.

$h=640+0.9\times2100=2530$ kJ/kg.

Exam tip: Given data → formula → substitution → answer with unit.

Conduction

Question: $k=0.8$, $A=10$, $L=0.2$, $\Delta T=30$.

$Q=0.8\times10\times30/0.2=1200$ W.

Exam tip: Given data → formula → substitution → answer with unit.

Practice and Quick Revision

One-line revision

- Gas law problems need absolute temperature.
- Area under P-V diagram represents work.
- Otto and Diesel cycles are air-standard cycles.
- Steam table use depends on steam condition.
- Heat transfer mode decides formula.

Industrial notes

- Boiler safety and water level are critical.
- Compressor overheating indicates lubrication/cooling/load issue.
- Heat exchanger fouling lowers performance.
- IC engine performance needs accurate fuel, speed and load measurement.

Model Paper and Complete Answers

Time: 3 Hours

Max Marks: 100

Part A

1. Define thermodynamic system.

2. State Zeroth law.
3. Write pV=mRT.
4. Define air-standard efficiency.
5. Distinguish SI and CI engines.
6. Define BP and IP.
7. Define dryness fraction.
8. Name boiler mountings.
9. State Fourier law.
10. Define heat exchanger effectiveness.

Part B / C

11. Explain thermodynamic systems.
12. Derive gas equation and Mayer relation.
13. Explain thermodynamic processes with P-V diagrams.
14. Explain first and second laws.
15. Draw and explain Otto and Diesel cycles.
16. Explain petrol and diesel engine working.
17. Explain valve timing and port timing.
18. Explain Morse test and heat balance sheet.
19. Explain steam formation and steam table use.
20. Explain Cochran or Babcock boiler.
21. Compare impulse and reaction turbine.
22. Solve plane wall conduction problem.
23. Explain radiation terms.
24. Compare heat exchangers.
25. Explain single-stage reciprocating compressor.

Answer guide

For long answers write definition, labelled diagram, working/derivation, formula, application and common mistake. For numericals write given data, formula, substitution, result and unit.

References

Official references

- Thermal Engineering - D. S. Kumar

- A Text Book of Thermal Engineering - R. S. Khurmi & J. K. Gupta
- Thermal Engineering - P. L. Ballany
- Elements of Heat Engines Vol I & II - R. C. Patel & C. J. Karamchandani
- Fundamentals of Thermal-Fluid Sciences - Yunus A. Cengel
- Elements of Mechanical Engineering - Sadhu Singh

Source limitation

Official information is taken from the supplied syllabus. Missing data is not fabricated.